HAMDARD UNIVERSITY KARACHI · TUESDAY, 10 NOVEMBER 2026

PROMPTOBIA

Explore. Build. Reflect.

Ethical AI · NCCIA Katalyst · Prompt Engineering with Antigravity · A Vibe-Coded Medical Agent

1 Ethical AI

2 Katalyst

3 Prompting

4 Medical Agent

5 Build & Reflect

Host: Iqbal Uddin Khan

Department of IT, FEST · Auditorium, FEST · 10:00 AM – 1:00 PM

EVENT AGENDA

Five sessions, three hours

1	Ethical Use of AI	10:00–10:30	30 min
2	NCCIA Katalyst Overview	10:30–11:00	30 min
3	Prompt Engineering with Antigravity	11:00–11:40	40 min
4	Hands-on Medical AI Agent	11:40–12:20	40 min
5	Build & Reflect	12:20–1:00	40 min

Event details

- Tuesday, 10 Nov 2026
- 10:00 AM – 1:00 PM
- Auditorium, FEST
- Host: Iqbal Uddin Khan
- Each session: short talk, task, Q&A

What you will leave with

1 Use AI responsibly

Value, risks, real cases, regulation and a checklist you can use tomorrow.

2 Understand Katalyst

A Pakistani example of AI for child protection, and the safety questions it raises.

3 Prompt with precision

Structure, tokens and Antigravity's plan-first workflow.

4 Build and evaluate

A vibe-coded medical agent that checks its own answers, plus your own project.

1

Real value, when used well

1 Productivity

Drafts, summaries, code and study aids in seconds, freeing time for judgment and creativity.

2 Health and science

Faster image review, literature search and triage support, with experts making the decisions.

3 Public services

Better forecasting and scheduling for energy, transport and large caseloads.

Value only counts if the output is trustworthy. The AI drafts; you stay accountable for what is used, shared or shipped.

1

Five pillars of responsible AI

1. Fairness and equity: test for bias across different groups
2. Transparency: explain how an output was produced and what data it used
3. Accountability: a named person or organization answers for results
4. Privacy and security: protect personal data, secrets and access
5. Safety and oversight: people can review, override and stop the system

1

Risks and the guardrail for each

Hallucination	Verify claims against trusted sources
Bias	Test outputs across groups; review regularly
Privacy	Never paste personal or patient data
IP and licensing	Check licenses and permissions; credit sources
Over-reliance	Keep a human in the loop for important decisions

Medical content raises the stakes: educational information only, never diagnosis or dosing.

1

Air Canada's chatbot (2024)

What happened

- Feb 2024: a Canadian tribunal ruled in Moffatt v. Air Canada
- The chatbot gave wrong bereavement-fare advice; a passenger relied on it
- Air Canada argued the chatbot was a separate entity; the tribunal rejected this
- Award: about C$812 in total (C$650.88 damages, plus interest and fees)

Lessons

- You own what your chatbot says
- Check answers against the real policy
- Small award, big signal: a tribunal decision, not binding elsewhere

1

The ChatGPT 'Sky' voice (2024)

What happened

- May 2024: OpenAI paused its 'Sky' voice after actress Scarlett Johansson said it sounded 'eerily similar' to hers
- She said she had declined an earlier offer to voice the system
- OpenAI said Sky used a different actress's natural voice and was not an imitation
- The dispute was public and contested; no court ruling

Lessons

- Get consent for voice and likeness
- Document how synthetic voices are made
- Perception matters as much as intent

1

The New York Times v. OpenAI

Where it stands (Nov 2026)

- Filed Dec 2023 against OpenAI and Microsoft
- The Times says its articles were copied to train models; defendants argue fair use
- 2025: the judge let the main copyright claims proceed
- Sept 2026: summary-judgment motions filed; no trial date; no final ruling

Lessons

- Training-data permission is still an open legal question
- Track the source and license of data
- Allegations are not findings

1

Gemini people images (2024)

What happened

- Feb 2024: Google paused Gemini's image generation of people after historically inaccurate images
- Google said its tuning to show a range of people failed to account for cases that should not show a range
- The model also became more cautious than intended

Lessons

- Bias fixes need context and testing
- Test in both directions
- Roll out in stages; be ready to pause

ADDED CONTEXT

1

EU AI Act: dates and penalties

2 Feb 2025	Prohibited practices apply (for example social scoring)
2 Aug 2025	Rules for general-purpose AI models apply
2 Dec 2026	New bans (abusive imagery) and content-marking rules for older systems
2 Dec 2027	Stand-alone high-risk systems (hiring, education, infrastructure)
2 Aug 2028	High-risk AI inside regulated products

7%

of global turnover, or EUR 35M, maximum fine for prohibited practices

Dates reflect the 2026 Digital Omnibus amendment. Verify before relying.

Four steps to put ethics into practice

1

1 Assign ownership

An ethics committee or named owner defines policy and approves high-risk uses.

2 Red-team before launch

Test with adversarial and edge-case inputs, including bias and misuse.

3 Keep humans in the loop

Human review for critical actions; clear ways to override and stop.

4 Audit continuously

Log outputs, monitor drift and incidents, and review the rules regularly.

Your responsible-use checklist

1. Before: remove personal and patient data
2. During: be specific; ask for sources
3. After: verify against two trusted sources
4. Sharing: disclose AI help; check licenses
5. Always: you are accountable for the result

Quick tasks (4 min)

- Redact: strip private details from a sample prompt (2 min)
- Verify: check one AI claim against two sources (2 min)

ADDED CONTEXT

1

AI-enabled cyber attacks: the evidence

~50

organizations hit by one agent-swarm operation

<6 hrs

to build and run an agent-enabled credential campaign

31% / 13%

breaches via exploited flaws vs stolen credentials

Not yet

fully autonomous attacks confirmed in the wild

AI changed the economics of attacks (speed, scale, skill needed), not the basic kinds of attack. Humans still choose targets and cash out; fundamentals still decide who gets breached.

ADDED CONTEXT

1

Myths vs realities: the top five

Common claim	Verdict	What the evidence shows
AI hackers are fully autonomous	Mostly myth	Humans set targets and review; models make mistakes
AI created new kinds of attack	Mostly myth	Familiar methods, cheaper and faster (NCSC, Anthropic)
80% of ransomware uses AI	Myth	The source paper was withdrawn after expert criticism
Only nation-states can do this	Myth	Lone operators ran multi-victim campaigns
AI makes basic security obsolete	Myth	Exploited flaws and stolen credentials still dominate

ADDED CONTEXT

1

Defender playbook: process over detection

Patch exploited flaws first

Shrink edge exposure

Phishing-resistant MFA

Restrict device-code flows

Out-of-band checks

Call back before paying

Secrets hygiene

Scan, vault, rotate

Treat AI keys as production credentials

Spend limits and alerts

Least agency for AI agents

Minimal tools; human approval

Deepfake lesson (Arup, 2024): a convincing face or voice is not proof of identity.

Ethics: four imperatives

1

1. Accept the duty of care: you answer for what your AI says and does
2. Audit data and licenses: know where training and input data came from
3. Balance guardrails: avoid both harm and over-correction; test both ways
4. Institutionalize review: keep people accountable for high-stakes decisions

Discussion: Where should AI stop and a human decide?

ADDED CONTEXT

2

What Katalyst is, and its uses

Definition

An AI-enabled investigative tool from Pakistan's NCCIA that helps detect, prioritize and investigate online child sexual exploitation and abuse.

Uses

- Triage large volumes of referrals
- Rank urgent cases first
- Scale analyst review capacity
- Support investigators with organized evidence

1,200 → 100,000+

NCMEC referrals NCCIA can review per month, as stated at launch

Launched in Islamabad on 28 Jan 2026 with the UK-funded UPSCALE program.

ADDED CONTEXT

2

How AI triage fits the workflow

Referrals arrive

Reports from tech platforms via NCMEC

→

AI triage

Rank cases so urgent harm is seen first

→

Analyst review

A human decides what is real and urgent

→

Investigation

Evidence handling and legal process

→

Action

Protect the child; pursue offenders

Typical building blocks in such tools (general, not confirmed for Katalyst)

- Matching against databases of known material
- Classifiers that flag new material for review
- Case management with audit logging and chain of custody

ADDED CONTEXT

2

Capabilities and safety considerations

1 **Detect**

Surface reports and evidence that need attention

2 **Prioritize**

Rank cases so urgent harm is reviewed first

3 **Investigate**

Support analysts with organized evidence

Safety considerations

- Humans make every decision; AI only assists
- Access limited to authorized staff, with audit logs
- Verifiable chain of custody for court use
- Limit analysts' exposure to harmful material
- No real case material in demos or training
- Review false positives and bias regularly

ADDED CONTEXT

2

Why it matters in Pakistan

~1M

NCMEC referrals reaching Pakistan each year

150,542

cybercrime complaints to NCCIA in 2025, mostly financial fraud

14-20 yrs

prison term for child exploitation after PECA amendments (2025)

Report online harm to the NCCIA helpline: 1799. Katalyst supports the agency's work; it does not replace reporting, schools, parents or the courts.

ADDED CONTEXT

2

Limits and open questions

1 New and synthetic media

Matching finds known files; new or AI-generated material needs classifiers and human review.

2 Errors and bias

False positives and negatives can harm people; needs testing, audit and appeals.

3 Cross-border limits

Privacy and data-sharing laws constrain how fast intelligence can move.

4 Transparency

Public detail is limited; independent evaluation and oversight build trust.

ADDED CONTEXT

2

Alternatives and peer tools

Tool	What it does	Main user
AviaTor	Triages NCMEC reports; installed locally	Law enforcement
Griffeye Analyze	Investigation platform; Project VIC certified	Law enforcement
Cellebrite triage	AI-assisted triage and automation	Law enforcement
S21 LASERi-X	CSAM investigation with shared intelligence data	Law enforcement
Thorn Safer	Detect, review and report CSAM on platforms	Tech companies

Katalyst is NCCIA's own deployment. Compare scope, data access and human oversight, not only features.

ADDED CONTEXT 2

Short tasks: think it through

1 Define · 2 min

Write a one-sentence definition of Katalyst for a non-technical friend.

2 Safeguard · 3 min

List three controls you would demand before trusting such a tool.

3 Compare · 2 min

Pick the closest peer tool from the alternatives slide and say why.

4 Fact-check · 3 min

Ask an AI to explain AI triage in 100 words; check it against the definition slide.

Katalyst: responsible AI, high stakes

2

1. AI speeds up triage; people make the decisions
2. Safety controls matter as much as capability
3. Claims need sources: trace numbers to primary reports
4. Report online harm: NCCIA helpline 1799

Questions and discussion, then a short break before Prompt Engineering at 11:00 AM.

3

Three tiers of prompts

1 Conversational

Informal, broad intent, high ambiguity. Fine for brainstorming.

2 Directive

Explicit rules and constraints. A bounded task with a clear target.

3 Agentic

A goal-oriented spec: the AI plans, uses tools and works in steps.

```
1: Fix my app.
2: Fix the null-token error in auth.js; do not change the schema.
3: Build MedAgent; plan first; wait for my approval before coding.
```

3

How prompt quality shapes behavior

1 Context narrows guesswork

Stack, audience and goal tell the model what you mean.

2 Constraints bound the output

Limits and negative rules keep answers on target.

3 Examples fix the format

One or two samples beat a long description.

4 Provided facts ground answers

Give source text; the model invents less.

Structure reduces errors; it does not remove them. Always verify.

3

Structured vs unstructured (illustrative)

Unstructured

```
"Fix the authentication bug on my Node app."
```

- The model guesses stack details
- May rewrite more than needed
- Hours spent checking the output

Structured

```
"Context: Express + JWT.
Task: fix the null-token error in auth.js.
Constraint: do not change the schema."
```

- Targets the exact file and bug
- Preserves existing architecture
- Easier to review and verify

3

The ripple effect of ambiguity

1

Vague input

Missing technical context

→

2

Model fills gaps

It assumes parameters

→

3

Code drift

Incompatible patterns appear

→

4

Rework tax

Time lost fixing and re-prompting

Ten extra seconds spent on context often saves many minutes of fixing.

3

Prompt architecture cheat sheet

Part	Purpose	Example
Role	Set domain depth	Act as a senior systems engineer
Context	Define technical bounds	Stack: Next.js 14, PostgreSQL
Task	Give one clear action	Refactor the JWT validation logic
Constraints	Add negative guardrails	Do not alter the database schema
Format	Shape the output	Plan first, then files, then run steps
Examples	Show the pattern	Two input-output pairs

ADDED CONTEXT

3

Tokenization nuances

1 Case matters

'Python' and 'python' can become different tokens.

2 Punctuation matters

Colons, quotes and trailing marks change the tokens the model sees.

3 Delimiters help

Backticks, fences and tags separate code or data from instructions.

```
convert python array      -> casual, chat-like
Convert Python array:     -> reads as an instruction
`convert_python_array()`  -> marks a literal identifier
```

Illustrative: exact splits depend on each model's tokenizer. Signals, not guarantees.

3

Antigravity: review before execution

1

Intent prompt

You state the goal in natural language

2

Implementation plan

The agent proposes a plan and task list first

3

Human review

You read, comment on and approve the plan

4

Execution

The agent builds, then shows diffs and a walkthrough

In Planning mode the agent works like a junior engineer who shows the plan before touching files.

ADDED CONTEXT

3

Antigravity features that help prompts

1 Agent Manager and Editor

Delegate to agents or edit code yourself

2 Planning vs Fast mode

Plan first for complex work

3 Artifacts

Task list, plan, walkthrough, screenshots, recordings

4 Rules

Standing instructions for the workspace

5 Workflows and Skills

Saved prompts and reusable instruction sets

6 Permission policies

Choose how much the agent may do alone

Folder names and menus vary by version (Antigravity 2.0, May 2026). Check on the lab PCs.

3

Transforming AI-assisted development

Traditional AI chat

Prompt → Raw unverified code → Manual debugging → Tech debt

Plan-first agent workflow

Prompt → Plan → Your review → Build → Test and verify

The review step only works if you actually read the plan.

3

Weak prompt vs strong prompt

Weak

```
Make a medical chatbot app.
```

- No platform or stack
- No safety limits
- No output format
- Guesswork means rework

Strong

```
ROLE: Senior Electron engineer.
CONTEXT: Teaching demo; education only.
TASK: Build desktop app "MedAgent".
CONSTRAINTS: 2 APIs; keys in Settings;
  no diagnosis or dosing.
FORMAT: Plan, then files, then run steps.
```

3

How to write it: five steps

1	State the goal	Build a medical Q&A desktop app.
2	Add context	For students; education only.
3	Set limits and format	No dosing. Plan, files, steps.
4	Show an example	Here is one sample JSON score.
5	Critique, then refine	List 3 weaknesses, then fix them.

3

Prompt examples you can reuse

Few-shot

```
Rewrite as a one-line risk note.
"Pasted patient name" -> "Privacy: remove names."
"Unverified drug claim" ->
```

Plan first

```
Plan first, no code yet. List files, APIs and risks.
Wait for my "approved" before you write any code.
```

Critique

```
List 3 weaknesses in your answer, then rewrite it.
Keep it under 150 words and end with a disclaimer.
```

3

Rules and workflows in practice

RULE FILE (standing instructions)

```
# .agent/rules/medagent.md
- Electron + plain JS
- Never hard-code API keys
- End answers with a
  medical disclaimer
- Plan first; ask before
  installing packages
```

WORKFLOW FILE (saved prompt)

```
# .agent/workflows/test-agent.md
# run with /test-agent
Run the app. Ask 3 safe and
2 red-flag questions.
Report scores, loop count
and any error.
```

Rules are like system instructions; workflows are saved prompts. Folder names vary by version.

3

Task: improve a weak prompt (6 min)

1	**Run the weak prompt**	Use 'Make a medical chatbot app.' in Planning mode
2	**Note the gaps**	Stack, safety limits, format, packaging
3	**Rewrite it**	Role, context, task, constraints, format
4	**Compare the plans**	Weak plan and strong plan side by side
5	**Name the winner**	Which added line changed the result most?

SUMMARY

Precision prompts, better results

3

1. Structure over length: clear parts beat long rambling
2. Punctuation and delimiters signal how to read the input
3. Plan before code: review the plan, then execute
4. Verify the output: structure reduces errors; it does not remove them

Next at 11:40 AM: build a medical agent with two APIs.

4

Two APIs: generate, then evaluate

User question → API A Generate → API B Evaluate (JSON) → API B Refine, max 2× → Safe answer + scores

1 LLM judge

Scores accuracy, safety, completeness and clarity from 1 to 5.

2 Pass rule

Average ≥ 4 and no safety flag; otherwise refine and re-check.

3 Scope

Education only: no diagnosis, no dosing, urgent-care advice for red flags.

4

The vibe-coding prompt

```
ROLE: Senior Electron + Node.js engineer.
GOAL: Desktop app "MedAgent": educational medical Q&A, NOT diagnosis.
FLOW: 1) User asks. 2) API A drafts an answer. 3) API B scores it as JSON
  (accuracy, safety, completeness, clarity 1-5, flags). 4) If avg < 4 or a
  safety flag, API B rewrites using its critique (max 2 loops). 5) Show
final
  answer, scores, loop count and token usage.
CONSTRAINTS: Keys and model names live in a Settings screen, never in
code.
  Call APIs from the main process. Red-flag symptoms: advise urgent care.
  Always show a medical disclaimer.
UI: one window, question box, answer card, score bars.
OUTPUT: Plan first, then files, then exact run and build steps
(.deb, .exe).
```

4

Development flow in Antigravity

1 Set up

Create workspace; add rules file

2 Prompt

Paste master prompt in Planning mode

3 Review plan

Check files, APIs, safety steps

4 Approve

Agent scaffolds and codes the app

5 Run & test

npm start; try safe and red-flag questions

6 Iterate

Fix with small, specific prompts

4

Compile: .deb and Windows installer

LINUX → .deb

```
npm install
npm run dist:linux
sudo apt install \
  ./dist/medagent_1.0.0_amd64.deb
medagent
```

WINDOWS → .exe installer

```
npm install
npm run dist:win
dist\MedAgent Setup 1.0.0.exe
```

electron-builder reads the "build" block in package.json: linux target deb (add rpm if needed; it requires rpmbuild), win target nsis. Build each installer on its own OS or in CI, and add a 512×512 icon in build/.

4

Push to a blank Git repository

```
git init -b main
printf
"node_modules/\ndist/\n.env\n" > .gitignore
git add .
git commit -m "feat: MedAgent v1.0.0"
git remote add origin \
  https://github.com/<user>/medagent.git
git push -u origin main
```

Before you push

- Repo is blank: no README or license
- No API keys in any file
- Installers go to Releases, not Git
- Sign in via token or gh CLI

ADDED CONTEXT

4

Session run sheet: 40 minutes

11:40–11:45	Demo the finished agent	5 min
11:45–11:55	Paste the prompt; review the plan together	10 min
11:55–12:10	Approve, build, run, fix with small prompts	15 min
12:10–12:20	Test: safe and red-flag questions; show scores	10 min

Keep installers and a finished repo ready as a backup if the live build stalls.

5

Your turn: build, test, reflect

1. Define your app idea in one sentence
2. Write a spec prompt (role to format)
3. Generate in Antigravity; review the plan
4. Run it and test with the team checklist
5. Reflect: which prompt change helped most?

ADDED CONTEXT

Time plan

- 12:20 Brief (5 min)
- 12:25 Build (20 min)
- 12:45 Test with checklist (10 min)
- 12:55 Reflect and wrap up (5 min)

ADDED CONTEXT

5

Tokens, cost and prompt quality

1 Tokens, not words

Models read tokens: roughly 4 English characters each. Other scripts can use more.

2 Length costs

Longer prompts cost more and slow replies; keep specs short and precise.

3 Vague costs more

Vague prompts cause rework, which spends more tokens overall.

4 Measure it

MedAgent shows token usage per question. Compare before and after a prompt change.

ADDED CONTEXT

5

What the team will test

Check	What to verify	Pts
Functionality	App runs; a question gets an answer	20
Prompt quality	Role, task, constraints, format present	15
Two-API loop	Generate, score, refine all work	20
Safety	Disclaimer; red flags; no dosing	15
Tokens & cost	Usage shown; prompt trimmed	10
Packaging	Installer builds and installs	10
Repo hygiene	Blank-repo push; no keys	10

Total 100 points. The weights are a suggestion; adjust to your grading scheme.

5

Reflect: what did you learn?

1. Which single prompt change improved your result the most?
2. Where did the evaluator disagree with the generator, and who was right?
3. What would you test next to make the agent safer?
4. What would you never let an AI do without a human check?

Share one lesson with the room: a prompt, a mistake or a surprise.

SUMMARY

Five takeaways for the day

1. AI drafts, humans decide: you stay accountable
2. Claims need sources: trace numbers and stories to primary reports
3. Prompts are specifications: structure beats length
4. Plan, build, then evaluate before you trust
5. Explore. Build. Reflect. Keep going after today

Key references

- APP, Express Tribune, Dawn: NCCIA launches Katalyst (Jan to Feb 2026)
- Moffatt v. Air Canada, 2024 BCCRT 149 (ABA Business Law Today, Feb 2024)
- NPR, AP: OpenAI pauses 'Sky' voice (20 and 21 May 2024)
- Google blog: Gemini image generation issue (Raghavan, 23 Feb 2024)
- New York Times v. OpenAI: AFP, NPR, case trackers (2025 to 2026)
- EU Digital Omnibus on AI: Taylor Wessing, Mondaq (May to July 2026)
- Google Antigravity guides and field notes (2025 to 2026)
- Anthropic threat report (10 Sept 2026); Google GTIG (8 Sept 2026); Verizon DBIR 2026
- UK NCSC: Impact of AI on cyber threat to 2027 (May 2025)
- Full list with URLs: Step-by-Step Guide V3, Appendix B

EXPLORE · BUILD · REFLECT

Questions?

1. AI drafts, humans decide
2. Prompts are specifications
3. Evaluate before you trust

Thank you · Host: Iqbal Uddin Khan · PROMPTOBIA · Hamdard University Karachi